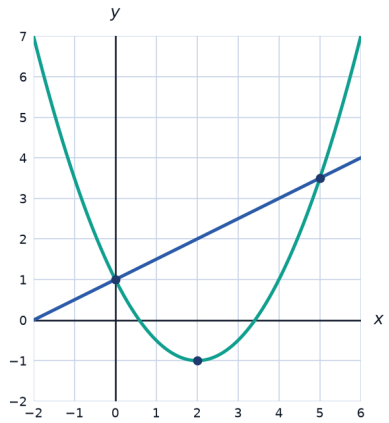
Regions defined by inequalities (Extended)

Core Concepts: Regions defined by inequalities (Extended)

1 For the model $f(x) = 2x + 7$ in regions defined by inequalities (extended), evaluate $f(2)$.



a. 11

b. 13

c. 4

d. 15

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for regions defined by inequalities (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 2x + 7$ in regions defined by inequalities (extended), evaluate $f(3)$.

a. 13

b. 15

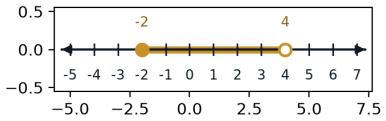
c. 6

d. 17

5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$



c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

6 A data set recorded for regions defined by inequalities (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 2x + 7$ in regions defined by inequalities (extended), evaluate $f(4)$.

a. 15

b. 17

c. 8

d. 19

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8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

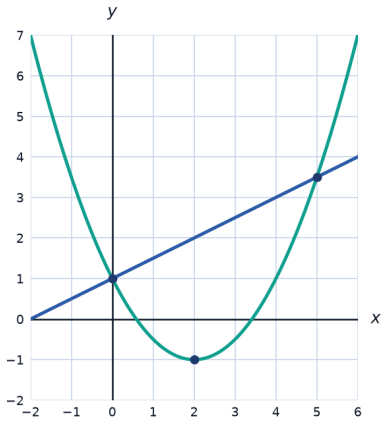
a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

9 A data set recorded for regions defined by inequalities (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?



a. 16.5

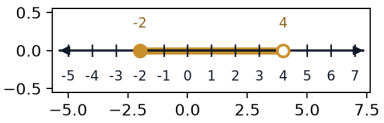
b. 17.5

c. 18.5

d. 20.5

Exam-Style Problem Solving & Modeling

10 For the model $f(x) = 2x + 7$ in regions defined by inequalities (extended), evaluate $f(5)$.



a. 17

b. 19

c. 10

d. 21

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11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for regions defined by inequalities (extended) consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

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